

Volatility-Aware SPY Strategy Backtest

A Python-Based Risk Control Strategy Report

1. Introduction

The purpose of this report is to evaluate whether a volatility-aware trading strategy can improve risk control when applied to SPY, an exchange-traded fund that tracks the S&P 500 Index. Instead of attempting to predict future market prices, this project uses historical market data to test how a simple rules-based strategy would have performed in the past.

The strategy is compared against a buy-and-hold benchmark, where SPY is purchased at the beginning of the backtest period and held continuously. The volatility-aware strategy adjusts market exposure based on recent market volatility: when short-term volatility is higher than its recent average level, the strategy moves out of SPY and holds cash; otherwise, it remains invested in SPY.

The main research question of this report is whether reducing exposure during high-volatility periods can lower downside risk compared with a simple buy-and-hold approach. The analysis focuses on key performance metrics including annualized return, annualized volatility, Sharpe ratio, and maximum drawdown.

2. Data and Methodology

This project uses historical daily price data for SPY from January 2023 to June 2026. The data was collected using Python through the 'yfinance' package, and the analysis was conducted using 'pandas', 'numpy', and 'matplotlib'.

The main price variable used in this report is the daily closing price of SPY. Based on the closing price, daily returns were calculated as the percentage change from one trading day to the next. A 20-day rolling volatility measure was then calculated using the standard deviation of daily returns over the previous 20 trading days. To create a relative volatility benchmark, the 60-day average of the 20-day rolling volatility was also calculated.

Rows with missing values were removed after the rolling calculations. The final dataset includes daily observations with closing price, daily return, 20-day rolling volatility, 60-day average volatility, trading signal, strategy return, and cumulative return.

3. Strategy Rules

Two strategies are tested in this report. The first strategy is a buy-and-hold benchmark. Under this approach, SPY is purchased at the beginning of the backtest period and held continuously until the end of the period. This benchmark is used to evaluate whether the volatility-aware strategy provides any improvement in risk-adjusted performance.

The second strategy is the volatility-aware strategy. This strategy uses recent market volatility to determine whether to stay invested in SPY or move out of the market. Specifically, the strategy compares the 20-day rolling volatility with the 60-day average of the 20-day rolling volatility. If the 20-day rolling volatility is less than or equal to the 60-day average volatility, the strategy holds SPY. If the 20-day rolling volatility is higher than the 60-day average volatility, the strategy holds cash instead.

A trading signal is created based on this rule. A signal value of 1 means the strategy is invested in SPY, while a signal value of 0 means the strategy is out of SPY and holding cash. To avoid look-ahead bias, the strategy return is calculated using the previous day's signal rather than the same day's signal.

4. Backtest Results

Figure 1. SPY Price with High-Volatility Periods

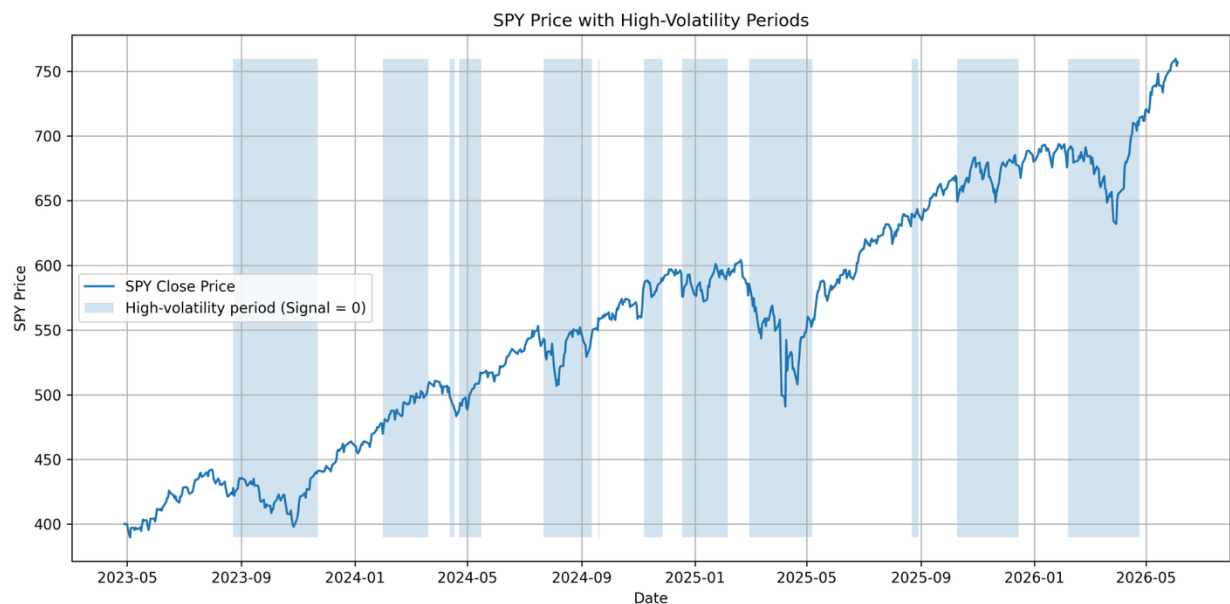


Figure 1 shows the closing price of SPY over the backtest period, with shaded areas representing high-volatility periods. These shaded regions occur when the 20-day rolling volatility is higher than its 60-day average. During these periods, the volatility-aware strategy moves out of SPY and holds cash. This helps illustrate how the strategy reduces market exposure when recent volatility is elevated.

Figure 2. SPY Buy-and-Hold vs Volatility-Aware Strategy

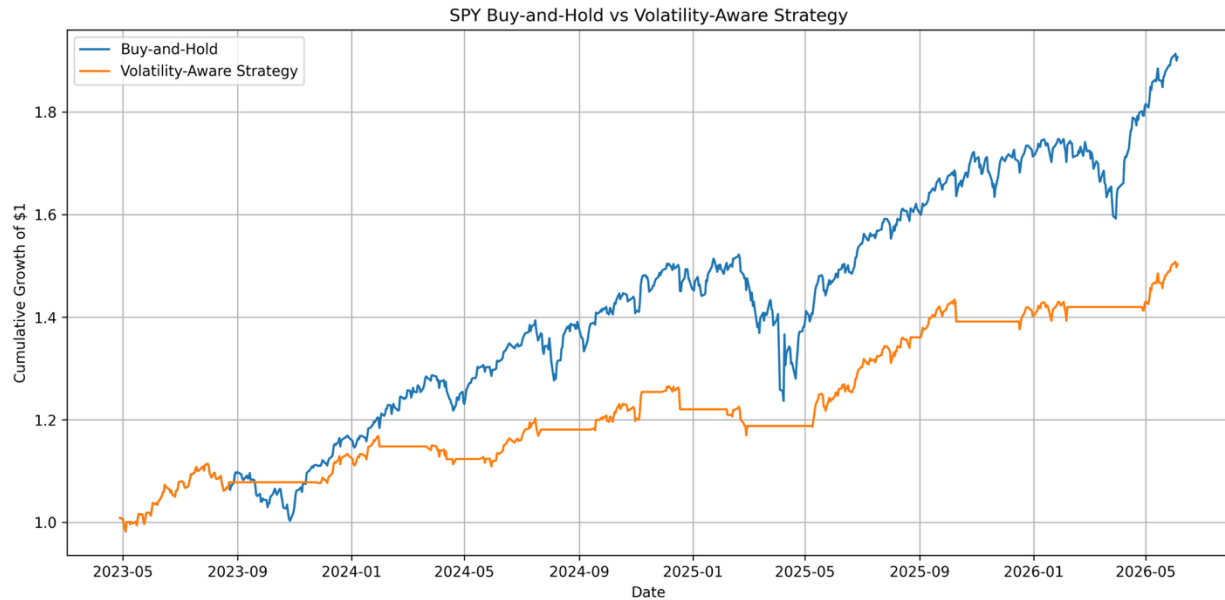


Figure 2 compares the cumulative growth of \$1 invested using the buy-and-hold strategy and the volatility-aware strategy. The buy-and-hold strategy produced higher cumulative growth over the backtest period. However, the volatility-aware strategy experienced a smoother return path because it reduced exposure during periods of higher market volatility.

Table 1. Performance Metrics Comparison

Metric	Buy-and-Hold	Volatility-Aware Strategy
Annualized Return	23.27%	14.12%
Annualized Volatility	15.02%	8.19%
Sharpe Ratio	1.55	1.72
Max Drawdown	-18.76%	-7.55%

The performance metrics provide a more detailed comparison between the two strategies. The buy-and-hold strategy achieved a higher annualized return of 23.27%, compared with 14.12% for the volatility-aware strategy. However, the volatility-aware strategy had a lower annualized volatility of 8.19%, compared with 15.02% for buy-and-hold. The volatility-aware strategy also achieved a higher Sharpe ratio of 1.72, compared with 1.55 for buy-and-hold, suggesting stronger risk-adjusted performance.

The largest difference appears in maximum drawdown. The buy-and-hold strategy had a maximum drawdown of -18.76%, while the volatility-aware strategy had a maximum drawdown of only -7.55%. This suggests that the volatility-aware strategy was more effective at controlling downside risk, even though it generated a lower total return.

5. AI-Assisted Interpretation

An AI-assisted interpretation was generated using the OpenAI API based on the backtest performance metrics.

Buy-and-Hold delivered the higher annualized return (23.27% vs. 14.12%), but it did so with substantially higher volatility (15.02% vs. 8.19%). The Volatility-Aware Strategy produced a higher Sharpe ratio (1.72 vs. 1.55), indicating better risk-adjusted performance despite its lower raw return.

The Volatility-Aware Strategy also experienced a much smaller maximum drawdown (-7.55% vs. -18.76%), suggesting a smoother ride and tighter risk control. Overall, the Volatility-Aware Strategy is better suited for risk control, while Buy-and-Hold maximized returns in this backtest.

6. Limitations

This backtest has several limitations. First, the strategy is based only on historical SPY price data and does not guarantee future performance. Market conditions can change, and a strategy that performed well in one period may not perform the same way in the future.

Second, the strategy does not include transaction costs, bid-ask spreads, slippage, taxes, or other real-world trading frictions. Since the volatility-aware strategy may move in and out of SPY multiple times, these costs could reduce the actual performance if the strategy were implemented in a real trading environment.

Third, the strategy uses a simple rule based on 20-day rolling volatility and its 60-day average. This rule is easy to understand, but it may not capture all important market information. For example, the strategy does not use macroeconomic indicators, interest rates, earnings data, news sentiment, or other signals that may affect market movements.

Finally, the analysis is limited to SPY over one specific historical period. Testing the strategy on other assets, longer time periods, or different market environments would be necessary before drawing stronger conclusions.

7. Conclusion

This report tested a volatility-aware strategy for SPY and compared it with a simple buy-and-hold benchmark. The results show that buy-and-hold produced a higher annualized return over the backtest period. However, the volatility-aware strategy reduced annualized volatility and maximum drawdown, while also achieving a higher Sharpe ratio.

These results suggest that the volatility-aware strategy may be more effective as a risk-control approach than as a return-maximization strategy. By moving out of SPY during periods of elevated volatility, the strategy reduced downside risk and created a smoother return path. However, this risk reduction came

at the cost of lower total returns, especially during periods when the market recovered quickly after volatility increased.

Overall, the strategy demonstrates how a simple volatility filter can be used to adjust market exposure and manage risk. Future improvements could include testing different volatility windows, adding transaction costs, using additional market indicators, or applying the strategy to other ETFs and asset classes.

The analysis was implemented in Python, and the backtest outputs include the cleaned dataset, performance metrics, generated figures, and an AI-assisted interpretation produced through the OpenAI API.